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Localized IQ mismatch measurement and compensation systems and methods

Abstract

The present disclosure relates to quadrature (IQ) signal mismatch in RFDACs. IQ mismatch is a non-ideality which occurs whenever I and Q vectors deployed to modulate the phase and the amplitude of RFDAC output signals are not aligned to be offset 90 degrees between each other. IQ mismatch may result in EVM degradation, and may be especially problematic for the RFDAC architecture because the RFDAC operates its digital-to-analog conversion at RF rate. A system may be deployed in the RFDAC to measure and compensate for IQ mismatch. The system may enable full-CMOS digital integration of both the measurement and compensation procedures. The system uses a simple flip-flop structure and exploits the fact that phase-vectors in CMOS logic are eventually digital pulses, and as such the mismatch between I and Q pulses with identical rise-fall time is simply measured as a skewed rise-to-rise and fall-to-fall delay between phases.

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Background/Summary

BACKGROUND

(1) The present disclosure relates generally to pass devices and digital-to-analog converters (DACs).

(2) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the present techniques, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure.

(3) Numerous electronic devices—including televisions, portable phones, computers, wearable devices, vehicle dashboards, virtual-reality glasses, and more utilize—DACs to generate analog electrical signals from digitally coded data. For example, an electronic device may use one or more DACs to convert digital signals to analog signals for transmission via radio frequency (RF) circuitry. Additionally or alternatively, DACs may be used to drive pixels of an electronic display at specific voltages based on digitally coded image data to produce the specific luminance level outputs to display an image. In some scenarios, DACs may use quadrature signals (e.g., I and Q) for control and/or modulation, such as in the phase and/or amplitude modulation of RF transceiver signals generated by a DAC. However, IQ mismatch is a non-ideality that occurs whenever quadrature (I and Q) vectors deployed to modulate the phase and/or the amplitude of radio-frequency digital-to-analog converter (RFDAC) output signals are not aligned in a desired way (e.g., having a phase relationship other than 90 degrees). In some scenarios, IQ mismatch may affect error vector magnitude (EVM) performance of the RFDAC and reduce overall RFDAC performance.

SUMMARY

(4) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(5) In one embodiment, a digital-to-analog converter includes a delay line that includes a plurality of inverters selectively coupled to an output of the delay line via a first plurality of switches; an additive delay line that includes a second plurality of switches coupled to a plurality of capacitive elements, wherein the additive delay line is coupled between a first inverter and a second inverter of the plurality of inverters; and a latch sensor that includes an input coupled to the output of the delay line.

(6) In another embodiment, a system includes a delay line configured to receive a reference signal and apply one or more delays to the reference signal; a sensor coupled to an output of the delay line, the sensor configured to receive a first quadrature signal, and receive a second quadrature signal, and processing circuitry configured to determine a phase offset between the second

quadrature signal based on a first overlap between the first quadrature signal and the second quadrature signal.

(7) In yet another embodiment, a method includes determining, via a processor, a mismatch between a first phase of a first quadrature signal and a second phase of a second quadrature signal, applying, via the processor, a compensation to the first quadrature signal, the second quadrature signal, or both; determining, via the processor, an error in the compensation; and adjusting, via the processor, the compensation to mitigate the error.

(8) Various refinements of the features noted above may exist in relation to various aspects of the present disclosure. Further features may also be incorporated in these various aspects as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to one or more of the illustrated embodiments may be incorporated into any of the above-described aspects of the present disclosure alone or in any combination. The brief summary presented above is intended only to familiarize the reader with certain aspects and contexts of embodiments of the present disclosure without limitation to the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various aspects of this disclosure may be better understood upon reading the following detailed description and upon reference to the drawings described below in which like numerals refer to like parts.

(2) FIG. 1 is a block diagram of an electronic device, according to embodiments of the present disclosure;

(3) FIG. 2 is a schematic diagram of the electronic device of FIG. 1, according to embodiments of the present disclosure;

(4) FIG. 3 is a schematic diagram of a transmitter of the electronic device of FIG. 1, according to embodiments of the present disclosure;

(5) FIG. 4 is a schematic diagram of a portion of the electronic device of FIG. 1 including a digital-to-analog converter (DAC) of the transmitter of FIG. 3, in accordance with an embodiment of the present disclosure;

(6) FIG. 5 is a flowchart of a method for converting a digital signal to an analog signal using the DAC of FIG. 4, in accordance with an embodiment of the present disclosure;

(7) FIG. 6 is a schematic diagram of a fractal DAC, in accordance with an embodiment of the present disclosure;

(8) FIG. 7 is a schematic diagram of a decision unit of the fractal DAC of FIG. 6, in accordance with an embodiment of the present disclosure;

(9) FIG. 8 is a schematic diagram of a column and line DAC, in accordance with an embodiment of the present disclosure;

(10) FIG. 9 is a graph illustrating IQ (e.g., quadrature signal) mismatch associated with the DAC of FIG. 4 and/or the fractal DAC of FIG. 6;

(11) FIG. 10 is a diagram illustrating the measurement of the I signals and the Q signals with respect to a reference signal, in accordance with an embodiment of the present disclosure;

(12) FIG. 11 is schematic diagram of a flip-flop sensor that may determine the overlap between the reference signal and I signal/Q signal to determine a change of state of the I signal and/or the Q signal;

(13) FIG. 12 is a schematic diagram of a tapped delay line that may provide delay to the reference signal to enable measurement of the I signal and the Q signal;

(14) FIG. 13 is a schematic diagram of an additive delay circuit that may provide an additional or

additive delay to the reference signal delay provided by the tapped delay line, according to embodiments of the present disclosure;

(15) FIG. 14 is a schematic diagram of an additive delay metal-oxide semiconductor (MOS) circuit that may serve as an alternative implementation of the additive delay circuit of FIG. 13, according to embodiments of the present disclosure;

(16) FIG. 15 is an illustration of various signal sequences used to measure the rising and falling edges of the I signal and the Q signal, according to embodiments of the present disclosure;

(17) FIG. 16 is a flowchart of a method for measuring the edge positions of the I signal and Q signal via the reference signal and the measurement circuitry, according to embodiments of the present disclosure;

(18) FIG. 17 is an illustration of fast logic signal sequences that may serve as an alternative to the various signal sequences of FIG. 15, according to embodiments of the present disclosure;

(19) FIG. 18 is a diagram of a selection circuit that enables selection of phases of the reference signal and flip-flop sensor signals (e.g., readback signals), according to embodiments of the present disclosure;

(20) FIG. 19 is a block diagram of a self-calibration loop for compensating IQ mismatch, according to embodiments of the present disclosure;

(21) FIG. 20 is a schematic diagram of the analog compensation circuit, according to embodiments of the present disclosure;

(22) FIG. 21 is a flowchart of a method for applying analog compensation correction to mitigate or eliminate IQ mismatch compensation error, according to embodiments of the present disclosure; and

(23) FIG. 22 is a flowchart of a method for determining an IQ mismatch and compensating for the IQ mismatch, according to embodiments of the present disclosure.

DETAILED DESCRIPTION OF SPECIFIC EMBODIMENTS

(24) One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(25) When introducing elements of various embodiments of the present disclosure, the articles “a,” “an,” and “the” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Additionally, it should be understood that references to “one embodiment” or “an embodiment” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, the particular features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. Use of the terms “approximately,” “near,” “about,” “close to,” and/or “substantially” should be understood to mean including close to a target (e.g., design, value, amount), such as within a margin of any suitable or contemplable error (e.g., within 0.1% of a target, within 1% of a target, within 5% of a target, within 10% of a target, within 25% of a target, and so on). Moreover, it should be understood that any exact values, numbers, measurements, and so on, provided herein, are contemplated to include approximations (e.g., within a margin of suitable or contemplable error) of the exact values, numbers, measurements, and so on.

(26) The present disclosure relates to digital-to-analog conversion. In particular, the present

disclosure relates to measuring and mitigating quadrature (IQ) signal mismatch in digital-to-analog converters (DACs) (e.g., having a column-and-line or fractal layout) such as those used in radio frequency (RF) communications (e.g., cellular communications). IQ mismatch is a non-ideality that occurs whenever the I and Q vectors (e.g., deployed to modulate the phase and the amplitude of RFDAC output signals) are not aligned as desired (e.g., offset by other than 90 degrees from each other). In general, quadrature (IQ) signal mismatch non-ideality may affect performance characteristics of the RFDAC IQ architecture when transmitting a modulated signal. For example, IQ mismatch may result in error vector magnitude (EVM) degradation, which may be exasperated at higher frequencies, and may be further exasperated in cases where the RFDAC performs digital-to-analog conversions at the RF rate associated with the IQ signals. As should be appreciated although discussed herein in the context of RFDACs, the techniques discussed herein may be utilized to measure and/or compensate for IQ mismatch in any electronic device that utilizes quadrature signals.

(27) As discussed herein, a system may be deployed in the RFDAC to measure and compensate for IQ mismatch. In some embodiments, the system may be implemented via CMOS digital integration for the measurement and/or compensation procedures. For example, the system may use a latch (e.g., flip-flop) circuit and exploit the fact that phase-vectors in CMOS logic are may be measured as digital pulses. As such the mismatch between I and Q pulses may be measured as a skewed rise-to-rise and fall-to-fall delay (e.g., in, milliseconds (ms), microseconds (μ s), picoseconds (ps), etc.) between phases.

(28) With the foregoing in mind, FIG. 1 is a block diagram of an electronic device **10**, according to embodiments of the present disclosure. The electronic device **10** may include, among other things, one or more processors **12** (collectively referred to herein as a single processor for convenience, which may be implemented in any suitable form of processing circuitry), memory **14**, nonvolatile storage **16**, a display **18**, input structures **20**, an input/output (I/O) interface **22**, a network interface **24**, and a power source **26**. The various functional blocks shown in FIG. 1 may include hardware elements (including circuitry), software elements (including machine-executable instructions) or a combination of both hardware and software elements (which may be referred to as logic). The processor **12**, memory **14**, the nonvolatile storage **16**, the display **18**, the input structures **20**, the input/output (I/O) interface **22**, the network interface **24**, and/or the power source **26** may each be communicatively coupled directly or indirectly (e.g., through or via another component, a communication bus, a network) to one another to transmit and/or receive data between one another. It should be noted that FIG. 1 is merely one example of a particular implementation and is intended to illustrate the types of components that may be present in electronic device **10**.

(29) By way of example, the electronic device **10** may include any suitable computing device, including a desktop or notebook computer, a portable electronic or handheld electronic device such as a wireless electronic device or smartphone, a tablet, a wearable electronic device, and other similar devices. In additional or alternative embodiments, the electronic device **10** may include an access point, such as a base station, a router (e.g., a wireless or Wi-Fi router), a hub, a switch, and so on. It should be noted that the processor **12** and other related items in FIG. 1 may be generally referred to herein as “data processing circuitry.” Such data processing circuitry may be embodied wholly or in part as software, hardware, or both. Furthermore, the processor **12** and other related items in FIG. 1 may be a single contained processing module or may be incorporated wholly or partially within any of the other elements within the electronic device **10**. The processor **12** may be implemented with any combination of general-purpose microprocessors, microcontrollers, digital signal processors (DSPs), field programmable gate array (FPGAs), programmable logic devices (PLDs), controllers, state machines, gated logic, discrete hardware components, dedicated hardware finite state machines, or any other suitable entities that may perform calculations or other manipulations of information. The processors **12** may include one or more application processors, one or more baseband processors, or both, and perform the various functions described herein.

(30) In the electronic device **10** of FIG. 1, the processor **12** may be operably coupled with a memory **14** and a nonvolatile storage **16** to perform various algorithms. Such programs or instructions executed by the processor **12** may be stored in any suitable article of manufacture that includes one or more tangible, computer-readable media. The tangible, computer-readable media may include the memory **14** and/or the nonvolatile storage **16**, individually or collectively, to store the instructions or routines. The memory **14** and the nonvolatile storage **16** may include any suitable articles of manufacture for storing data and executable instructions, such as random-access memory, read-only memory, rewritable flash memory, hard drives, and optical discs. In addition, programs (e.g., an operating system) encoded on such a computer program product may also include instructions that may be executed by the processor **12** to enable the electronic device **10** to provide various functionalities.

(31) In certain embodiments, the display **18** may facilitate users to view images generated on the electronic device **10**. In some embodiments, the display **18** may include a touch screen, which may facilitate user interaction with a user interface of the electronic device **10**. Furthermore, it should be appreciated that, in some embodiments, the display **18** may include one or more liquid crystal displays (LCDs), light-emitting diode (LED) displays, organic light-emitting diode (OLED) displays, active-matrix organic light-emitting diode (AMOLED) displays, or some combination of these and/or other display technologies.

(32) The input structures **20** of the electronic device **10** may enable a user to interact with the electronic device **10** (e.g., pressing a button to increase or decrease a volume level). The I/O interface **22** may enable electronic device **10** to interface with various other electronic devices, as may the network interface **24**. In some embodiments, the I/O interface **22** may include an I/O port for a hardwired connection for charging and/or content manipulation using a standard connector and protocol, such as the Lightning connector provided by Apple Inc. of Cupertino, California, a universal serial bus (USB), or other similar connector and protocol. The network interface **24** may include, for example, one or more interfaces for a personal area network (PAN), such as an ultra-wideband (UWB) or a BLUETOOTH® network, for a local area network (LAN) or wireless local area network (WLAN), such as a network employing one of the IEEE 802.11x family of protocols (e.g., WI-FI®), and/or for a wide area network (WAN), such as any standards related to the Third Generation Partnership Project (3GPP), including, for example, a 3rd generation (3G) cellular network, universal mobile telecommunication system (UMTS), 4th generation (4G) cellular network, long term evolution (LTE®) cellular network, long term evolution license assisted access (LTE-LAA) cellular network, 5th generation (5G) cellular network, and/or New Radio (NR) cellular network, a satellite network, and so on. In particular, the network interface **24** may include, for example, one or more interfaces for using a Release-15 cellular communication standard of the 5G specifications that include the millimeter wave (mmWave) frequency range (e.g., 22.25-300 gigahertz (GHz)) and/or any other cellular communication standard release (e.g., Release-16, Release-17, any future releases) that define and/or enable frequency ranges used for wireless communication. The network interface **24** of the electronic device **10** may allow communication over the aforementioned networks (e.g., 5G, Wi-Fi, LTE-LAA, and so forth).

(33) The network interface **24** may also include one or more interfaces for, for example, broadband fixed wireless access networks (e.g., WIMAX®), mobile broadband Wireless networks (mobile WIMAX®), asynchronous digital subscriber lines (e.g., ADSL, VDSL), digital video broadcasting-terrestrial (DVB-T®) network and its extension DVB Handheld (DVB-H®) network, ultra-wideband (UWB) network, alternating current (AC) power lines, and so forth.

(34) As illustrated, the network interface **24** may include a transceiver **28**. In some embodiments, all or portions of the transceiver **28** may be disposed within the processor **12**. The transceiver **28** may support transmission and receipt of various wireless signals via one or more antennas, and thus may include a transmitter and a receiver. The power source **26** of the electronic device **10** may include any suitable source of power, such as a rechargeable lithium polymer (Li-poly) battery

and/or an alternating current (AC) power converter. In certain embodiments, the electronic device **10** may take the form of a computer, a portable electronic device, a wearable electronic device, or other type of electronic device.

(35) FIG. **2** is a functional diagram of the electronic device **10** of FIG. **1**, according to embodiments of the present disclosure. As illustrated, the processor **12**, the memory **14**, the transceiver **28**, a transmitter **30**, a receiver **32**, and/or antennas **34** (illustrated as **34A-34N**, collectively referred to as an antenna **34**) may be communicatively coupled directly or indirectly (e.g., through or via another component, a communication bus, a network) to one another to transmit and/or receive data between one another.

(36) The electronic device **10** may include the transmitter **30** and/or the receiver **32** that respectively enable transmission and reception of data between the electronic device **10** and an external device via, for example, a network (e.g., including base stations) or a direct connection. As illustrated, the transmitter **30** and the receiver **32** may be combined into the transceiver **28**. The electronic device **10** may also have one or more antennas **34A-34N** electrically coupled to the transceiver **28**. The antennas **34A-34N** may be configured in an omnidirectional or directional configuration, in a single-beam, dual-beam, or multi-beam arrangement, and so on. Each antenna **34** may be associated with one or more beams and various configurations. In some embodiments, multiple antennas of the antennas **34A-34N** of an antenna group or module may be communicatively coupled to a respective transceiver **28** and each emit radio frequency signals that may constructively and/or destructively combine to form a beam. The electronic device **10** may include multiple transmitters, multiple receivers, multiple transceivers, and/or multiple antennas as suitable for various communication standards. In some embodiments, the transmitter **30** and the receiver **32** may transmit and receive information via other wired or wireline systems or means.

(37) As illustrated, the various components of the electronic device **10** may be coupled together by a bus system **36**. The bus system **36** may include a data bus, for example, as well as a power bus, a control signal bus, and a status signal bus, in addition to the data bus. The components of the electronic device **10** may be coupled together or accept or provide inputs to each other using some other mechanism.

(38) FIG. **3** is a schematic diagram of the transmitter **30** (e.g., transmit circuitry), according to embodiments of the present disclosure. As illustrated, the transmitter **30** may receive outgoing data **38** in the form of a digital signal to be transmitted via the one or more antennas **34**. A digital-to-analog converter (DAC) **40** of the transmitter **30** may convert the digital signal to an analog signal, and a modulator **42** may combine the converted analog signal with a carrier signal to generate a radio wave. Additionally or alternatively, the DAC **40** and modulator **42** may be implemented together in a DAC/modulator **44**. For example, the DAC/modulator **44** may convert the digital signal to the analog signal and combine the converted analog signal with the carrier signal simultaneously or concurrently and/or within the same circuitry. Moreover, the DAC/modulator **44** may be implemented as multiple circuits (e.g., DAC **40** and modulator **42**) coupled together or a singular combined circuit. In some embodiments, the DAC/modulator **44** may directly generate a modulated analog signal without first generating the converted analog signal. Furthermore, as used herein, a DAC **40** may refer to a standalone DAC **40** or a combined DAC/modulator **44**, and an analog signal may refer to a converted analog signal or a modulated analog signal. Additionally, while embodiments are described herein as applying to RF signal generation, in some embodiments, aspects of the present disclosure may be applicable to other types or utilizations of DACs, such as a baseband DAC.

(39) A power amplifier (PA) **46** receives the modulated signal from the modulator **42**. The power amplifier **46** may amplify the modulated signal to a suitable level to drive transmission of the signal via the one or more antennas **34**. A filter **48** (e.g., filter circuitry and/or software) of the transmitter **30** may then remove undesirable noise from the amplified signal to generate transmitted data **50** to be transmitted via the one or more antennas **34**. The filter **48** may include any suitable

filter or filters to remove the undesirable noise from the amplified signal, such as a bandpass filter, a bandstop filter, a low pass filter, a high pass filter, and/or a decimation filter. Additionally, the transmitter **30** may include any suitable additional components not shown, or may not include certain of the illustrated components, such that the transmitter **30** may transmit the outgoing data **38** via the one or more antennas **34**. For example, the transmitter **30** may include a mixer and/or a digital up converter. As another example, the transmitter **30** may not include the filter **48** if the power amplifier **46** outputs the amplified signal in or approximately in a desired frequency range (such that filtering of the amplified signal may be unnecessary).

(40) FIG. **4** is a schematic diagram of a portion of the transmitter **30** of the electronic device **10** having a DAC **40**, according to an embodiment of the present disclosure. In some embodiments, the DAC **40** may share a supply or positive power voltage (e.g., VDD) **52** provided by the power source **26** with other components **54** of the transmitter **30** and/or the electronic device **10**. For example, the other components **54** may include any powered electronic component of the transmitter **30** and/or the electronic device **10** utilizing the supply voltage **52** or a derivative thereof. Moreover, the DAC **40** may receive a digital signal **56** (e.g., of outgoing data **38**), an enable signal **58**, and/or a complementary enable signal **60**. The enable signal **58** and/or the complementary enable signal **60** may enable and/or facilitate enabling operation of the DAC **40**. For example, if the enable signal **58** is logically “low” relative to a reference voltage **62** (e.g., ground or other relative voltage), then the DAC **40** may be disabled or inactive (e.g., in a disable, inactive, or deactivated state). On the other hand, if the enable signal **58** is logically “high” (e.g., relative to the reference voltage **62** and/or the supply voltage **52**), then the DAC **40** may be enabled or active for operation (e.g., in an enabled or activated state). Furthermore, the reference voltage **62** (e.g., VSS) may be provided as a reference for the digital signal **56**, the enable signal **58**, the complementary enable signal **60**, the supply voltage **52**, and/or the analog output signal **64**. As should be appreciated, and as used herein, signals (e.g., the digital signal **56**, the enable signal **58**, the complementary enable signal **60**, the analog output signal **64**, etc.) may correspond to voltages and/or currents relative to a reference and may represent electronically storable, displayable, and/or transmittable data.

(41) As discussed herein, the different analog output signals **64** generated by the DAC **40** may correspond to values of the digital signal **56**. The digital signal **56** and corresponding analog output signal **64** may be associated with any suitable bit-depth depending on implementation. For example, in the context of image data (e.g., in a baseband DAC) and/or signal transmission data (e.g., in an RF DAC), an 8-bit digital signal **56** may correspond to 255 or 256 analog output signals **64**.

(42) FIG. **5** is a flowchart of a method **66** for converting a digital signal to an analog signal using the DAC **40**, according to an embodiment of the present disclosure. In general, the DAC **40** may receive a digital signal **56** representative of an analog signal (process block **70**). The DAC **40** may also generate an analog output signal **64** (as discussed in further detail below), utilizing power from the power source **26**, based on the received digital signal **56** (process block **80**). The generated analog output signal **64** may then be output from the DAC **40** (processing block **90**).

(43) As discussed above, the DAC **40** may generate an analog output signal **64** by enabling one or more unit cells to output a unit amount of current or voltage that, in the aggregate, form the analog output signal **64**. The unit current or voltage may be predetermined and based on implementation factors. For example, the unit cells may include one or more capacitors that store a fixed amount of charge that may be released to form the analog output signal **64**. In some scenarios, the physical and/or logical layout of the unit cells may affect the speed of operation of the DAC **40** and/or the linearity of the DAC **40**. As such, in some embodiments, one or more DACs **40** of the electronic device **10** may be implemented as a fractal DAC **100**, as illustrated in FIG. **6**. A fractal DAC **100** may include multiple unit cells **102** arranged (e.g., logically and/or physically) in a fractal pattern constructed of fractal blocks **104**. Moreover, the illustrated pattern may be replicated by replacing each unit cell **102** with a fractal block **104** to realize a fractal DAC of increased size while

maintaining symmetry.

(44) In the illustrated example, the fractal DAC **100** includes sixteen fractal blocks **104** of four unit cells **102**, which may correspond to, for example, sixty-four different analog output signals **64** (e.g., which may have non-zero values). However, larger fractal DACs may be envisioned by replacing each unit cell **102** with a fractal block **104**, increasing the size of the fractal DAC **100** by four each time to maintain $4 \times$ unit cells **102** (where x is the number of fractal block recursions in the fractal DAC **100**). As should be appreciated, the size of the fractal DAC **100** may depend on implementation factors such as desired granularity of the analog output signal **64**. Furthermore, different size fractal blocks **104** (e.g., half of a fractal block **104**) may be used to achieve different numbers of total unit cells **102** (e.g., $2 \times$ number of unit cells **102** for fractal blocks **104** having a size of two unit cells **102**). Moreover, in some embodiments, one or more unit cells **102** may be representative of fractional unit cells (e.g., outputting 0.5 or 0.25 of a unit voltage or current) to further increase granularity, dynamic range extension, and/or as an offset to decrease differential nonlinearity (DNL) and/or integral nonlinearity (INL).

(45) In some embodiments, the multiple nested fractal blocks **104** may be continuously/recursively split into symmetrical branches by decision units **106** (e.g., **106A**, **106B**, **106C**, **106D**, etc.) until reaching the unit cells **102**. That is, for a given branch of the fractal DAC **100**, sequential decision units **106** may be used to interpret and decode the digital signal **56** and direct enable/disable signals to the corresponding unit cells **102** to generate the analog output signal **64**. Additionally, although the digital signal **56** is depicted as a single line, in some embodiments, the digital signal **56** may include multiple data buses running in parallel through the fractal DAC **100**. For example, the multiple data buses may include data for multiple phases and/or polarity (e.g., negative and positive). As such, the fractal DAC **100** and the decision units **106** may operate using multiple digital signals **56** in parallel to control outputs of the unit cells **102**.

(46) To help illustrate, FIG. 7 is an example decision unit **106** receiving an incoming signal **108** of n bits, according to an embodiment of the present disclosure. In some embodiments, the incoming signal **108** (e.g., the digital signal **56**) is a binary signal that is decoded step-by-step by the sequential decision units **106**, such that the aggregate of the signals reaching the unit cells **102** forms a thermometric signal. For example, the aggregate thermometric signal for a binary incoming signal **108** of “10” may be represented as “0011.” As the decision units **106** decipher and pass on certain portions of the incoming signal **108** along different routes, the unit cells **102** may eventually end up with respective portions of the thermometric digital signal (e.g., with logical “1” or high going to two unit cells **102** for activation and logical “0” or low going to two different unit cells **102** for deactivation). For example, the incoming signal **108** may have n -bits (e.g., abcdef . . . n , where each letter is representative of a logical value in a binary format, as in the illustrated example). Each decision unit **106** may take the most significant bit (MSb) of the incoming signal **108**, repeat it $n-1$ times, and output a MSb signal **110** having the MSb of the incoming signal **108** repeated $n-1$ times. Additionally, the decision unit **106** may output a least significant bit (LSb) signal **112** including the remainder of the incoming signal **108**, without the MSb, having $n-1$ total bits.

(47) As should be appreciated, the MSb of a binary signal is representative of half of the value of the incoming signal **108**. As such, if the MSb (e.g., at decision unit **106A**) is a logical “1”, the repeated logical “1” will be propagated down half of the branches of the fractal DAC **100**, reducing the bit-depth by one with each subsequent decision unit **106**, to enable half of the unit cells **102** downstream from the initial decision unit **106** (e.g., decision unit **106A**). The remaining half of the unit cells **102** may be enabled or disabled according to the LSb signal **112** having the remainder of the incoming signal **108**. Using similar logic, the LSb signal **112** from an initial decision unit **106** (e.g., decision unit **106A**) may be the incoming signal **108** for a subsequent decision unit **106** (e.g., decision unit **106B**) and so forth. Furthermore, while depicted as outputting the MSb signal **110** to the left and the LSb signal **112** to the right, decision units **106** may output the LSb signal **112** and

MSb signal **110** in either direction according to a fill order (e.g., an order of increasing activations of unit cells **102**) of the fractal DAC **100**, which may be programmable. Moreover, in some embodiments, a remainder bit may be added to the digital signal **56** prior to the fractal DAC **100** or added to the MSb signal **110** and/or LSb signal **112** at the first decision unit **106** (e.g., decision unit **106A**) based on the digital signal **56** to facilitate decoding from a binary digital signal to a thermometric digital signal (e.g., at the unit cells **102**).

(48) Additionally, although depicted in FIGS. **6** and **7** as having two outputs (e.g., MSb signal **110** and LSb signal **112**), in some embodiments, the decision units **106** may evaluate multiple bits of the incoming signal **108** at the same time (e.g., simultaneously or concurrently). For example, a decision unit **106** may provide four outputs in a quaternary split of the incoming signal **108**, effectively combining the efforts of the first two levels of decision units **106** (e.g., decision unit **106A**, decision unit **106B**, and the decision unit opposite decision unit **106B**). In the example of the quaternary split, two outputs may include the MSb signal **110** with a bit depth of $n-2$, a signal of repeated entries of the second MSb with a bit depth of $n-2$, and the LSb signal **112** with a bit depth of $n-2$, having the 2 MSBs removed. As should be appreciated, the number of splits for a single decision unit **106** may vary based on implementation. Furthermore, in some embodiments, the decision units **106** may include multiple incoming signals **108**, for example from multiple parallel data buses, and provide either a binary split, a quaternary split, or other split to each incoming signal **108**.

(49) As discussed above, the fractal DAC **100** may facilitate decoding of the digital signal **56** (e.g., via the decision units **106**) into a thermometric signal dispersed among the unit cells **102**. Additionally or alternatively, the digital signal **56** may include a binary signal that is not decoded via the decision units **106**. For example, some unit cells **102** may have a binary-sized output that is dependent upon a binary signal. In some embodiments, the binary signal (e.g., a portion of or separate from the digital signal **56**) may traverse the same path as the decoded thermometric signal and therefore have substantially similar arrival time at the binary coded unit cells **102**, maintaining synchronicity of the fractal DAC **100**. For example, the binary signal may be passed through or bypass the decision units **106** and/or use separate distribution logic following the data path of the fractal DAC **100**. The binary coded unit cells **102** may use the binary signal to vary the output between zero (e.g., disabled) and a full unit voltage or current (e.g., 0.0, 0.25, 0.5, 0.75, or 1.0 of a unit voltage or current). For example, the binary coded unit cell **102** may include binary interpretation logic to decode the binary signal and enable the binary coded unit cell **102** at an intermediate power level (e.g., 0.25, 0.5, or 0.75 of a unit voltage or current). The binary-sized output of the binary coded unit cells **102** may facilitate increasing resolution of the analog output signal **64** by providing increased granularity.

(50) In some embodiments, the DAC **40** or the DAC/modulator **44** may include a DAC other than the fractal DAC **100**, such as a column and line DAC **114** shown in FIG. **8**. In some scenarios, the column and line DAC **114** may include a multitude of control signals **116** from control logic **118** feeding an array of unit cells **102**. For example, the control logic **118** of the column and line DAC **114** may incorporate binary to thermometric conversion and/or take into consideration the desired states of multiple individual unit cells **102** concurrently or simultaneously to determine control signals **116** necessary for operation.

(51) The fractal DAC **100** may include data paths (physically and/or logically) to each unit cell **102** that are substantially of the same dimensions, components, and/or number of components, which may further increase linearity and/or synchronicity. For example, returning briefly to FIG. **6**, starting from the incoming digital signal **56** and the first decision unit **106A**, the data path to each unit cell **102** and the number of decision units **106** traversed along the data path is the same for each unit cell **102**. In other words, a fractal DAC **100** may include data paths that are substantially the same, innately providing the decoded incoming signal **108** to each of the unit cells **102** concurrently or at substantially the same time. As should be appreciated, in some embodiments,

some data paths of a fractal DAC **100** may differ due to manufacturing tolerances, physical layout constraints, data-line-to-data-line coupling, and/or additional implementation factors and interference. The substantially similar data paths of the fractal DAC **100** may reduce or eliminate a wait time associated with the difference between shorter and longer data paths (e.g., the difference between data path **120** and data path **122** in a column and line DAC **114**), further increasing the operable speed of the fractal DAC **100**. As discussed above, the decision units **106** may recursively split the digital signal **56**, at each level of decision unit(s) **106**, and output an MSb signal **110** and an LSb signal **112** to different branches of the fractal DAC **100**. As used herein, the “level” of a decision unit **106** may refer to how many decision units **106** have been traversed by the digital signal **56**. For example, referring back to FIG. **6**, decision unit **106A** may be considered to be at level one, decision unit **106B** may be considered to be at level two, and so on.

(52) As discussed above, the techniques discussed herein may be applicable to a fractal DAC **100**, a column and line DAC **114**, or any suitable device that uses quadrature signals. Furthermore, as should be appreciated, while discussed herein as utilized with respect to quadrature signals (e.g., signals with 90-degree phase relationships), the present techniques may also be applicable to measure and/or compensate for phase mismatches in any suitable set of phasor-related signals having a set offset. With this in mind, FIG. **9** is a graph **150** illustrating IQ (e.g., quadrature signal) mismatch. IQ mismatch is a non-ideality that occurs whenever the I and Q vectors of the quadrature signals are not aligned to be offset 90 degrees from each other. An I signal **152** and a Q signal **154** have no mismatch when there is a 90-degree offset between the phase of the I signal **152** and the phase of the Q signal **154**. However, if the phase of the I signal **152** or the Q signal **154** is such that the offset between the I signal **152** and the Q signal **154** is greater or less than 90 degrees, such as is shown by the I' signal **156**, there exists an IQ mismatch **158**.

(53) The I signal **152** and the Q signal **154** may, in some cases, be measured via direct or indirect measurement during manufacturing (e.g., after fabrication). For example, an output signal of an RFDAC utilizing a quadrature signal as a local oscillator (LO) may be measured, and an IQ mismatch **158** derived therefrom. In some cases, IQ mismatch may be determined once during manufacturing by generating RF waves (e.g., a transmitted signal), measuring an error with respect to the transmitted signal, and setting/applying a pre-distortion (e.g., a digital pre-distortion) to the operating signals of the RFDAC. However, in some cases IQ mismatch may increase or change over time or during normal operation of an electronic device (e.g., due to various device non-idealities such as age and environment). Consequently, it may be beneficial to measure and correct for IQ mismatch **158** throughout the life of the electronic device **10** (e.g., periodically or continuously during operation of the electronic device **10**). In doing so, IQ mismatch due to non-idealities (e.g., heat, age of the electronic device **10**) may be mitigated or eliminated during normal use and operation of the electronic device **10**.

(54) As previously stated, at certain frequencies (e.g., RF rate frequencies), IQ mismatch **158** may affect the EVM performance of the RFDAC IQ architecture when transmitting a modulated signal. Indeed, IQ mismatch **158** may result in EVM degradation, which may impact the overall RFDAC performance. As such, in some embodiments, IQ compensation circuitry may be deployed in the RFDAC to measure and compensate for the IQ mismatch **158**. The IQ compensation circuitry may enable digital integration (e.g., full-CMOS integration) of the measurement and/or compensation procedures. In some embodiments, the IQ compensation circuitry uses a flip-flop structure to exploit the fact that phase-vectors in CMOS logic include digital pulses, and, as such, the mismatch between I and Q pulses with may be measured as a skewed rise-to-rise and fall-to-fall delay (e.g., in picoseconds (ps)) between phases. While the signal mismatch measuring and compensation is discussed below with respect to quadrature signals (e.g., I signals **152** and Q signals **154**), it should be noted that such measurement and compensation may apply to any signals with a desired phase relationship.

(55) To perform accurate mitigation measures, accurate measurement of the rising and falling edges

of the I signals **152** and Q signals **154** may be determined. FIG. **10** is a diagram illustrating the measurement of the I signals **152** and the Q signals **154** with respect to a reference signal, according to embodiments of the present disclosure. The Q signals **154A**, **154B**, and **154C** (collectively, the Q signals **154**) and the I signals **152A**, **152B**, and **152C** (collectively, the I signals **152**) may include digital pulses that may be measured (e.g., in the time domain) with respect to reference signals **202A**, **202B**, **202C**, **202D**, **202E**, and **202F** (collectively, the reference signal **202**). The Q signals **154A**, **154B**, and **154C** may include pulses of the Q signal **154** at various times and the I signals **152A**, **152B**, and **152C** may include pulses of the I signal **152** at various times. (56) The reference signal **202** may be deployed with various delays to enable accurate measurement of the rising and falling edges of the Q signal **154** and the I signal **152**. For example, the reference signals **202A** and **202D** may include a first delay (e.g., delay=0 ps), the reference signals **202B** and **202E** may include a second delay (e.g., delay=1 ps or more, 2 ps or more, 5 ps or more, 10 ps or more, and so on), and the reference signals **202C** and **202F** may include a third delay (e.g., a delay=2 ps or more, 4 ps or more, 10 ps or more, 20 ps or more, and so on). As should be appreciated, the delays discussed herein may be of any suitable granularity, and the time associated with such delays may be based on implementation (e.g., frequency).

(57) The rising edge of the Q signal **154** and/or the I signal **152** may be measured based on the value of the delay applied to the reference signal **202**, when the reference signal **202** overlaps with the rising edge of the Q signal **154** and/or I signal **152**, which occurs at the point at which the pulse value of the Q signal **154** and/or the I signal **152** transitions from low to high. The position of the falling edge of the Q signal **154** and/or the I signal **152** may be measured based on the value of the delay applied to the reference signal **202** when the reference signal **202** overlaps with the falling edge of the Q signal **154** or I signal **152**, which occurs at the point at which the pulse value of the Q signal **154** and/or I signal **152** transitions from high to low.

(58) FIG. **11** is a schematic diagram of a latch sensor (e.g., flip-flop sensor **220**) that may be used to determine the overlap between the reference signal **202** and I signal **152**/Q signal **154** and to determine when a change of state of the I signal **152** and/or the Q signal **154** occurs relative to the reference signal **202**, according to embodiments of the present disclosure. When considering two pulses (e.g., pulses associated with reference signal **202** and the I signal **152** or the Q signal **154**), it may be beneficial to determine when the edges of the two pulses are early or late with respect to one another to determine if the reference signal **202** is detecting a rising or falling edge of the quadrature signals. The flip-flop sensor **220** may sample the reference signal **202** at the D input **222** using the Q signal **154** or the I signal **152** as a clock signal on the clock input **224**, which results in a high readback output (e.g., a binary 1) at the Q output **226** whenever the reference signal **202** is early with respect to the quadrature signal (e.g., indicating a rising edge of the quadrature signal) or results in a low readback output (e.g., a binary 0) at the Q output **226** whenever the reference signal **202** is late with respect to the quadrature signal (e.g., indicating a falling edge of the quadrature signal). It may be appreciated that the Q output **226** outputs a relative signal indicating the relationship between the reference signal **202** and a quadrature signal (e.g., the I signal **152** or the Q signal **154**).

(59) The compactness of the flip-flop sensor **220** allows for its placement at the core of the DAC **40**. In some embodiments, the flip-flop sensor **220** may be placed at or near the position of a unit cell **102** of the fractal DAC **100**, and may in some embodiments replace the unit cell **102**, ensuring that the IQ mismatch measured is as seen at the last output stage corresponding to the output of the IQ mismatch. Multiple sensors may be deployed, which may facilitate accounting for mismatch over multiple paths. For example, different paths or branches of the fractal DAC **100** and/or column and line DAC **114** may have different fabrication asymmetries, and deploying multiple flip-flop sensors **220** may enable the DAC **40** to account for multiple or all of the various asymmetries (e.g., and thus the various IQ mismatches). Multiple flip-flop sensors **220** may also provide enhanced functionality, such as measuring positive and negative edge positions simultaneously.

(60) It may be beneficial to deploy two flip-flop sensors **220**, one to measure the rising edge transition and one to measure the falling edge transitions of the quadrature signals, each with reversed clock polarity, respectively. In some embodiments, one flip-flop sensor **220** may be used. Indeed, one sensor is capable of detecting both rising and falling transitions of a signal by looking at the transition of the output **226** from 0 to 1 (quadrature signal rising edge) or 1 to 0 (quadrature signal falling edge). However, in some cases it may be difficult to determine the relationship between rising and falling edge positions (e.g., duty cycle). In such a configuration, a typical CMOS sensor such as a flip-flop may not have identical delay time between rising and falling transitions of an input signal. Consequently, it may be advantageous to use multiple flip-flop sensors **220** to measure rising and falling signal edges.

(61) FIG. **12** is a schematic diagram of a tapped delay line **250** that may provide delay to the reference signal **202** to enable measurement of the I signal **152** and the Q signal **154**. The tapped delay line **250** may include a set of cascaded inverters **252A**, **252B**, **252C**, **252D**, **252E**, and **252F** (collectively, the inverters **252**). The tapped delay line **250** may include switches coupled to tap points between the inverters **252**. For example, a switch **254A** is coupled to a tap point **256A** between the output of the inverter **252A** and the input of the inverter **252B**, a switch **254B** is coupled to a tap point **256B** between the output of the inverter **252C** and the input of the inverter **252D**, and a switch **254C** is coupled to tap point **256C** between the output of the inverter **252E** and the input of the inverter **252F**. The switches **254A**, **254B**, and **254C** may collectively be referred to herein as the switches **254**; and the tap points **256A**, **256B**, and **256C** may collectively be referred to herein as the tap points **256**. While the switches **254** are shown coupled to the tap points **256** between sets of two cascaded inverters **252**, it should be noted that the switches **254** may be coupled between any number of cascaded inverters **252** (e.g., between each inverter, between sets of three inverters or more, between sets of 10 cascaded inverters or more, and so on).

(62) The inverters **252** may each be associated with a unit of delay. For example, each inverter **252** may provide (e.g., in the analog domain) a delay of 1 picosecond (ps) or more, 5 ps or more, 20 ps or more, and so on to the reference signal **202**. The amount of delay in the tapped delay line **250** may be adjusted by selecting various switches **254**. For example, each inverter **252** may provide a 5 ps delay to the reference signal **202** in the tapped delay line **250** to generate delayed reference signal **258**. By closing the switch **254A**, a 5 ps delay may be provided to the reference signal **202** in the tapped delay line **250** to generate delayed reference signal **258**; by closing the switch **254B**, a 15 ps delay may be provided to the reference signal **202** in the tapped delay line **250** to generate delayed reference signal **258**; and by closing the switch **254C**, a 25 ps delay may be provided to the reference signal **202** in the tapped delay line **250** to generate/provide the delayed reference signal **258**.

(63) By closing one of the switches **254** and adjusting the delay of the tapped delay line **250**, different delays associated with the reference signal **202** may be probed until a transition region (e.g., indicating a transition from high-to-low or low-to-high) associated with the I signal **152** and/or the Q signal **154** may be identified. To detect the change of state indicating a transition region, a sensor (e.g., the flip-flop sensor **220**) may be used as described with respect to FIG. **11**. Accordingly, the tapped delay line **250** may enable an amount of delay applied to the reference signal **202** in an accurately controlled manner to determine the position of the reference signal **202** with respect to the I signal **152** and the Q signal **154**.

(64) By determining a delay step of the tapped delay line **250** (e.g., if the delay step includes a 10 ps delay step) and determining an overlap between the reference signal **202** and the quadrature signals (e.g., the I signal **152** and/or the Q signal **154**), as discussed with respect to FIG. **10**, the distance between the I signal **152** and the Q signal **154** (e.g., the IQ mismatch) may be determined. For example, briefly returning to FIG. **10**, by determining that the delay step is 10 ps it may be estimated that the I signal **152** and the Q signal **154** are at a distance of 1 code (e.g., 10 ps), and that the pulse width of the I signal **152** and the Q signal **154** is 20 ps. If it is determined that the

frequency of operation includes a 40 ps period, the I signal **152** and the Q signal **154** may be determined to be in the desired quadrature relationship (e.g., offset from each other by 90-degrees). (65) It should be noted that since relative distances are used to determine the phase relationship between the pulses, the absolute value of the measurement itself may not be considered. If, for example, the absolute position of the edge of the I signal **152** would measure to be 2.6 code instead of 2.5 code, it may be determined that the IQ mismatch includes a misalignment of 0.1 code (e.g., 1 ps). Therefore, correct estimate of the IQ mismatch may be provided by direct pulse edge position measurement.

(66) It may be appreciated that, in some cases, accuracy is given by the delay step fineness (e.g., 10 ps) in the measurement. However, depending on implementation (e.g., frequency) the tapped delay line **250** may be relatively coarse in relation to the IQ mismatch **158**. Indeed, in some scenarios <1-degree error may correspond to sub-picosecond accuracy (e.g., at certain (higher) frequencies). Accuracy of the measurement may be augmented by means of averaging in the presence of randomness. Indeed, by performing multiple measurements with a degree of random error, the systematic measurement of the IQ mismatch **158** may be averaged for increased accuracy. Adopting, for example, a 10 ps step size for the tapped delay line **250** (with no or negligible non-linearity error) and targeting an error of 0.5 ps (e.g., 0.9-degree error at 5 gigahertz (GHz)), such accuracy may be achieved with roughly 500 randomized measurement points. In some embodiments, a counter may be used to keep track of the number of randomized measurement points to ensure desired accuracy.

(67) When having a differential non-linearity (DNL) superposed to the 10 ps step size such as that the delay line steps are irregularly spaced (e.g., such as a sequence of steps including 9 ps, 11 ps, 10 ps), the measurement procedure may not be limited by DNL performance as long as averaging occurs over multiple delay elements of the tapped delay line **250**. This may be achieved by means of applying a randomization method which is strong enough in its impact to have an edge position occur over multiple codes of the tapped delay line **250** when measuring (e.g., a randomization greater than the time step of the tapped delay line **250**), which may average the residual DNL.

(68) In some instances, it may be beneficial to adjust the delay of the reference signal **202** in intervals that are greater than (e.g., more coarse-grain) or smaller than (e.g., more fine-grain) than the delay provided by the tapped delay line **250**. FIG. **13** is a schematic diagram of an additive delay circuit **300** that may provide an additional or additive delay to the reference signal delay provided by the tapped delay line **250**, according to embodiments of the present disclosure. The additive delay circuit **300** may include a switch **302A**, a switch **302B**, and a switch **302C** (referred to herein as the switches **302**) coupled between the inverters **252A** and **252B** (e.g., of the tapped delay line **250**).

(69) Each of the switches **302** may be coupled to a capacitive element, such as a capacitor, such that a capacitor **304A** is coupled between the switch **302A** and a ground **306**, a capacitor **304B** is coupled between the switch **302B** and the ground **306**, and a capacitor **304C** is coupled between the switch **302C** and the ground **306**. By closing one or more of the switches **302**, various delays may be applied to the reference signal **202**. For example, by closing the switch **302A**, a delay of 1 ps may be applied to the reference signal **202**, by closing the switch **302B**, a delay of 2 ps may be applied to the reference signal **202**, and by closing the switch **302C**, a delay of 4 ps may be applied to the reference signal **202C**. As should be appreciated, the additive delay circuit **300** may be implemented before, after, or between inverters of the tapped delay line **250**.

(70) FIG. **14** is a schematic diagram of an additive delay metal-oxide semiconductor (MOS) circuit **350** that may serve as an alternative implementation of the additive delay circuit **300** discussed with respect to FIG. **13**, according to embodiments of the present disclosure. The switches **302** of the additive delay circuit **300** may include two drain-source coupled transistors **352A** and **354A**. For example, the transistor **352A** may include an n-channel metal oxide semiconductor field-effect transistor (nMOS) and the transistor **354A** may include a p-channel metal oxide semiconductor

field-effect transistor (pMOS). The drain of the transistor 352A and the source of the transistor 354A are coupled together at the terminal 356. The source of the transistor 352A and the drain of the transistor 354A are coupled together at the terminal 358. The capacitors 304 of the additive delay circuit 300 may also be replaced by nMOS transistors. For instance, the source of the transistor 352A and the drain of the transistor 354A are coupled to a gate terminal of a transistor 360A at the terminal 358, and the drain and source terminals of the transistor 360A are coupled together (e.g., shorted) to the ground 306. The transistors 352B, 354B, and 360B and 352C, 354C, and 360C are also disposed in a similar arrangement to the transistors 352A, 354A, and 360A.

(71) Furthermore, in some embodiments, additional or alternative methods to generate an delays may be implemented. For example, a perturbation may be added to a VDD power of the tapped delay line 250 or the additive delay circuit 300/additive delay MOS circuit 350. A varied VDD may result in a different $V_{sub,TH}$ applied to the delay line gates and as such a different delay of the unit elements (e.g., different speed), where the higher the VDD the faster the delay line and the smaller the unit delay. When measuring rising and falling edges together, one error which may not be recoverable through averaging is the asymmetric behavior of the flip-flop sensor 220 and the tapped delay line 250 or the additive delay circuit 300/additive delay MOS circuit 350 with respect to rising and falling edges of the Q signal 154 and/or I signal 152. A double measurement circuit capable of detecting rising and falling edges simultaneously may be implemented in situations where no intrinsic offset occurs between the rising and falling edges measurement. Such a double measurement circuit may be identical to the measurement circuits described above with respect to FIGS. 10-14, with the polarity of the clock input 224 swapped such that the clock input 224 triggers in the opposite direction of the single measurement circuit described above. An accurate measurement of rising and falling edges relative positions enables the measurement circuits to calibrate IQ mismatch between phases of the Q signal 154 and the I signal 142, but also duty cycle errors within a single phase and between multiple phases.

(72) FIG. 15 is an illustration of various sequences of delays added to the reference signal 202 used to measure the rising and falling edges of the I signal 152 and the Q signal 154, according to embodiments of the present disclosure. To measure the position of the rising edges and the falling edges of the I signal 152 or the Q signal 154, a ramp on the tapped delay line programming may be provided to the reference signal 202, and the positions upon which changes to a readback signal (e.g., an output signal of the flip-flop sensor 220) may be stored as either rising or falling transition positions of the I signal 152 and/or the Q signal 154, depending on the direction of the transition itself. The tapped delay line programming ramp 402 illustrates delay line programming with respect to the reference signal 202, such as the delay provided by the tapped delay line 250 of FIG. 12. The additive delay programming ramp 404 illustrates additive delay, such as the additive delay provided by the additive delay circuit 300 or the additive delay MOS circuit 350. The readback ramp 406 illustrates readback, which includes a signal that is output from the flip-flop sensor 220. When delay line delay is increased, at a certain point the reference signal 202 crosses over (e.g., the reference signal 202E crosses over the I signal 152B, as illustrated in FIG. 10) and the readback becomes a logical high. The observed readback signal may include a series of logical highs (e.g., 1s) and logical lows (e.g., 0s). For example, if a signal has a period of 100 ps, and the tapped delay line 250 has a step of 10 ps, the signal may include a series of five 1s and five 0s, such that the signal may be represented as 1111100000.

(73) The pulses represented by the tapped delay line programming ramp 402, the additive delay programming ramp 404, and the readback ramp 406 may be averaged out to provide an accurate estimation of the rising and falling positions of the reference signal 202. To provide averaging functionality, the tapped delay line programming ramps 402 may be repeated per each one of the additive delay programming codes (e.g., a code including a 10 ps step size) on a synchronized ramp that may update once per delay line programming cycle. The first position of the edge's readback position are accumulated at every delay line programming cycle. At a certain time, an average edge

position may be determined after full sequencing is completed. The average edge position may be measured by an accumulator, which may accumulate all or a portion of all observed edge positions of the quadrature signal. The value determined by the accumulator may then be divided by the number of additive delay programming cycles.

(74) It may be observed from FIG. 15 that the tapped delay line programming ramp 402 includes several ascending ramps illustrating multiple delay line programming cycles, while the additive delay programming ramp 404 includes a descending ramp. Inverting the additive delay programming ramp 404 with respect to the tapped delay line programming ramp 402 may, together with deployment of anti-wrap-around logic, prevent the wrapping-around of signal edge position between subsequent delay line programming cycles. As discussed previously, the readback signals of the readback ramp 406 may include sequences of 1s and 0s. Consequently, it may become difficult to differentiate the end of a first signal and the beginning of a second signal at the next delay stage. For example, if a first signal pulse of a first delay line programming cycle ends in a series of 0s and a second (e.g., consecutive) signal pulse of a second delay line programming cycle begins in a series of 0s, it may be difficult to differentiate the delineation between the first and the second signal.

(75) When applying delay, due to the periodic nature of the observed pulses of the ramps, a delayed version of a reference pulse of the reference signal 202 may wrap around. That is, the first edge determined is not a delayed version of a first reference pulse of the reference signal 202, it may be an edge of a different pulse of the reference signal 202 entirely (e.g., a pulse corresponding to a subsequent delay line programming cycle).

(76) Inverting the additive delay programming ramp 404 with respect to the tapped delay line programming ramp 402 may substantially alter the step size between a last pulse of a first delay line programming cycle of the tapped delay line programming ramp 402 and a first pulse of a second delay line programming cycle of the tapped delay line programming ramp 402. Moreover, anti-wraparound logic may be implemented, which considers as first edges only those edges which are in proximity of a previously measured edge position. This may be implemented via a disabling pulse which masks readings which are too low (e.g., below a threshold) compared to the previously measured edge position (e.g., below a previous position diminished by a certain margin). Additionally, the inverse nature of the additive delay programming ramp 404 facilitates wraparound prevention by ensuring that the beginning of a subsequent pulse of the reference signal 202 is below the threshold.

(77) FIG. 16 is a flowchart of a method 450 for measuring the edge positions of the I signal 152 and Q signal 154 via the reference signal 202 and the measurement circuitry discussed above, according to embodiments of the present disclosure. Any suitable device (e.g., a controller) that may control components of the electronic device 10, may perform the method 450. In some embodiments, the method 450 may be implemented by executing instructions stored in a tangible, non-transitory, computer-readable medium, such as the memory 14 and/or storage 16, using the processor 12. For example, the method 450 may be performed at least in part by one or more software components, such as an operating system of the electronic device 10, one or more software applications of the electronic device 10, and the like. While the method 450 is described using steps in a specific sequence, it should be understood that the present disclosure contemplates that the described steps may be performed in different sequences than the sequence illustrated, and certain described steps may be skipped or not performed altogether.

(78) In process block 452, the processor 12 may measure a position of the reference signal 202 with respect to one or more quadrature signals (e.g., the I signal 152 and the Q signal 154). The processor 12 may perform the measurements using the processes and systems described above with respect to FIGS. 10-15. In process block 454, the processor 12 may provide delay via the tapped delay line 250, as discussed with respect to FIGS. 12-15. In process block 456, the processor may provide additive delay to the reference signal 202 via the additive delay circuit 300 (or the additive

delay MOS circuit **350**), as discussed with respect to FIGS. **13-15**.

(79) In query block **458**, the processor determines whether the reference signal **202** has overlapped with an edge of one or more quadrature signals (e.g., has not overlapped the edge of the I signal **152** and/or the Q signal **154**). In other words, it may be determined whether a transition has occurred to indicate a rising or falling edge of the I signal **152** and/or the Q signal **154**. If the processor **12** determines that the reference signal has not overlapped an edge of the one or more quadrature signals (e.g., as shown by the reference signal **202D** in FIG. **10**), in query block **460**, the processor **12** determines whether the reference signal **202** has reached the end of a delay line programming cycle. If the processor **12** determines in the query block **460** that the reference signal **202** has not reached the end of the delay line programming cycle, the processor may increment the delay provided to the reference signal via the tapped delay line **250**.

(80) However, if the processor **12** determines, in the query block **460**, that the reference signal **202** has reached the end of the delay line programming cycle, the processor may, in process block **462**, reset the delay line programming cycle (e.g., to begin a new delay line programming cycle), and return to the process block **456** to increment the additive delay provided to the reference signal **202** via the additive delay circuit **300** (or the additive delay MOS circuit **350**).

(81) Returning to the query block **458**, if the processor determines that the reference signal **202** does overlap with an edge of the one or more quadrature signals (e.g., as illustrated by reference signals **202A**, **202B**, **202C**, **202E**, and **202F** in FIG. **10**), the processor may, in process block **464**, store the edge position in the memory **14** or the storage **16**.

(82) In query block **466**, the processor determines whether an edge position counter has reached a measurement point counter limit. If the processor determines that the edge position counter has not reached a limit of the randomized measurement points, in process block **468** the processor may reset the programming of the delay lines and apply a delay via the tapped delay line **250** or the additive delay circuit **300**/additive delay MOS circuit **350**.

(83) However, if the processor determines that the edge position counter has reached the limit, in process block **470** the processor **12** averages the edge measurements to determine the relative positions of the I signal **152** and the Q signal **154**. Based on the determination of the positions of the I signal **152** and Q signal **154**, the processor may determine the IQ mismatch **158**. In this manner, the method **450** may enable measurement of the edge positions of the I signal **152** and Q signal **154** via the reference signal **202** and the various delay and measurement circuitries previously described.

(84) FIG. **17** is an illustration of logic signal sequences that may serve as an alternative to the various signal sequences discussed with respect to FIG. **15**, according to embodiments of the present disclosure. The logic sequences may be implemented to provide anti-wraparound functionality with added speed benefits. For example, if an edge position of the one or more quadrature signals is found during a first delay line programming cycle, the additive delay programming may apply a smaller change to narrow the search range of positions sampled in subsequent programming cycles.

(85) The search range of the delay line programming cycle may be restricted after the first cycle completes, an initial position of the edge of a quadrature signal is known, to an interval around the initially discovered edge position. The interval may be defined to have at its lowest the first discovered position diminished by a margin M and at its highest the discovered position augmented by a margin μ for a total length of L . The range might be dynamically updated after every cycle upon a newly found edge position, or assumed to be valid throughout the sequence. As may be appreciated from fast logic delay line programming **502**, fast logic additive delay programming **504**, and fast logic readback, by restricting the range of cycles subsequent to the first cycle in which a quadrature signal edge is found, the sequence may execute faster than without the use of the fast logic.

(86) FIG. **18** is a diagram of a selection circuit **550** that enables selection of phases of the reference

signal **202** and flip-flop sensor **220** signals (e.g., readback signals), according to embodiments of the present disclosure. The selection circuit may enable comparison of four phases: Q phase (Q.sub.p) **552**, I phase (I.sub.p) **554**, Q-not (Q.sub.n) **556**, and I-not (I.sub.n) **558**. Q-not **556** is the inverse of the Q phase **552** and I-not **558** is the inverse of the I phase **554**.

(87) To compare the IQ mismatch between the four phases shown, the selection circuit **550** may be utilized, enabling switching of any selected phase to the same sensor (e.g., the flip-flop sensor **220**). Using one sensor may enable comparison of the phase alignments without introducing sensor-to-sensor mismatches, as opposed to if multiple sensors were deployed for each phase. Additionally, to improve the averaging capabilities, the selection circuit **550** may be instantiated both at the source-side of the reference signal **202** (e.g., before the tapped delay line **250** and the additional delay circuit **300**/additional delay MOS circuit **350**), and also before the flip-flop sensor **220** measurement terminal. This placement may enable construction of an IQ mismatch matrix, which may improve accuracy of the measurement results.

(88) Recalling that the flip-flop sensor **220** is a two-input device with a D input **222** and a clock input **224** (e.g., as discussed with respect to FIG. **11**), the reference selection switches **560A**, **560B**, **560C**, and **560D** (collectively, the reference selection switches **560**) may select any of the phases as an input signal to the D input **222**. The sensor selection switches **562A**, **562B**, **562C**, and **562D** may select any of the phases to serve as the clock at the clock input **224**. Accordingly, as there are four phases Q phase **552**, I phase **554**, Q-not **556** and I-not **558**, there may be generated a 16-entry phase comparison matrix to compare the phase alignments between any two of the phases. Such a matrix may enable fast and accurate phase calculation to determine IQ mismatch.

(89) It should be noted that the four phases illustrated in the selection circuit **550** are fully available to the consumer of the phases regardless of if a calibration sequence is running or not.

Consequently, background calibration of the phases is enabled by the selection circuit **550**.

(90) FIG. **19** is a block diagram of a self-calibration loop **600** for compensating IQ mismatch, according to embodiments of the present disclosure. The self-calibration loop includes an analog compensation circuit **602**, sensor hardware **604** (e.g., the flip-flop sensor **220**, the delay circuitry, and so on), and an analog circuit **606** (e.g., a circuit that may receive the quadrature signals via the DAC **40** or the fractal DAC **100**). Due to the large number of calculations that may be performed in identifying the edge positions of the I signal **152**, the Q signal **154** and the overall IQ mismatch, a finite state machine **608** may be implemented to increase the speed of the edge detection sequencing. As the calculations and matrix manipulation described above may consume substantial memory, firmware **610** may be used for performing calculation (e.g., using a microprocessor, a controller, and so on).

(91) FIG. **20** is a schematic diagram of the analog compensation circuit **602**, according to embodiments of the present disclosure. The analog compensation circuit **602** may enable compensation of IQ mismatch with independent rise and fall transition control. The I signal **152** and/or the Q signal **154** may enter the analog compensation circuit **602** via a source inverter **620**. The analog compensation circuit **602** includes a falling edge control component including an inverter **622**, falling edge delay circuitry **624**, and a pMOS **626**. The inverter **620**, the falling edge delay circuitry **624**, and a gate terminal of the pMOS **626** may be coupled together at a node **628**.

(92) The falling edge delay circuitry **624** includes switches **630A**, **630B**, and **630C** (collectively, the switches **630**) coupled to the node **628**. The falling edge delay circuitry **624** includes capacitive elements (e.g., capacitors, transistors, and so on) coupled to the switches **630** that enable various delays to be provided when coupled to the pMOS **626** via the switches **630**. The falling edge delay circuitry **624** includes a capacitor **632A** coupled between the switch **630A** and the ground **306**; a capacitor **632B** coupled between the switch **630B** and the ground **306**; and a capacitor **632C** coupled between the switch **630C** and the ground **306**. By closing one or more of the switches **630**, various delays may be applied to the falling edges of the I signal **152** and/or the Q signal **154**. For example, closing the switch **630A** may apply a delay of 1 ps to the falling edge of the I signal **152**.

and/or the Q signal **154**, closing the switch **630B** may apply a delay of 2 ps to the falling edge of the I signal **152** and/or the Q signal **154**, and closing the switch **630C** may apply a delay of 4 ps to the falling edge of the I signal **152** and/or the Q signal **154**.

(93) The analog compensation circuit **602** includes a rising edge control component including an inverter **640**, rising edge delay circuitry **642**, and an nMOS **644**. The inverter **640**, the rising edge delay circuitry **642**, and a gate terminal of the nMOS **644** are coupled to a node **646**. The rising edge delay circuitry **642** includes switches **648A**, **648B**, and **648C** (collectively, the switches **648**). The rising edge delay circuitry **642** includes capacitive elements (e.g., capacitors, transistors, and so on) coupled to the switches **648** that enable various delays to be provided when coupled to the nMOS **644** via the switches **648**. The falling edge delay circuitry **642** includes a capacitor **650A** coupled between the switch **648A** and the ground **306**; a capacitor **650B** coupled between the switch **648B** and the ground **306**; and a capacitor **650C** coupled between the switch **648C** and the ground **306**. By closing one or more of the switches **648**, various delays may be applied to the rising edges of the I signal **152** and/or the Q signal **154**. For example, closing the switch **648A** may apply a delay of 1 ps to the rising edge of the I signal **152** and/or the Q signal **154**, closing the switch **648B** may apply a delay of 2 ps to the rising edge of the I signal **152** and/or the Q signal **154**, and closing the switch **648C** may apply a delay of 4 ps to the rising edge of the I signal **152** and/or the Q signal **154**. The rising and/or falling edge adjustments supplied to the pMOS **626** and/or the nMOS **644** may be output to inverter **652**.

(94) In this manner, it should be appreciated that closing the switches **630**, the analog compensation circuit **602** may adjust the falling edges of the I signal **152** or the Q signal **154**; and by closing the switches **648**, the analog compensation circuit **602** may adjust the rising edges of the I signal **152** or the Q signal **154** to mitigate or eliminate IQ mismatch. It should be noted that, the switches **630** and **648** may be implemented as transistors (e.g., similar to the structure discussed with respect to the additive delay MOS circuit **350**). Further, while the capacitive elements are discussed as the capacitors **632** and **650**, the capacitive elements may include transistors as well (e.g., similar to the structure discussed with respect to the additive delay MOS circuit **350**). It should be noted that, while the analog compensation circuit **602** is discussed as including a rising edge control component and a falling edge control component separately, the analog compensation circuit **602** may be implemented with a single component (e.g., an inverter, delay circuitry, and MOS) that adjusts both the rising and falling edges of the I signal **152** and/or the Q signal **154**.

(95) In some cases, the IQ mismatch compensation provided by the analog compensation circuit **602** may include some compensation error, such that the IQ mismatch compensation does not compensate the IQ mismatch as desired. In such cases, it may be beneficial to evaluate and correct the IQ mismatch compensation provided by the analog compensation circuit **602**.

(96) FIG. **21** is a flowchart of a method **670** for applying analog compensation correction to mitigate or eliminate IQ mismatch compensation error, according to embodiments of the present disclosure. Any suitable device (e.g., a controller, the processor **12**, IQ mismatch compensation circuitry, and so on) that may control components of the electronic device **10**, may perform the method **670**. In some embodiments, the method **670** may be implemented by executing instructions stored in a tangible, non-transitory, computer-readable medium, such as the memory **14** and/or storage **16**, using the processor **12**. For example, the method **670** may be performed at least in part by one or more software components, such as an operating system of the electronic device **10**, one or more software applications of the electronic device **10**, and the like. While the method **670** is described using steps in a specific sequence, it should be understood that the present disclosure contemplates that the described steps may be performed in different sequences than the sequence illustrated, and certain described steps may be skipped or not performed altogether.

(97) In process block **672**, the processor **12** may measure a first IQ mismatch error when no mismatch compensation is applied via the analog compensation circuit **602**. In process block **674**, the processor **12** may measure a second IQ mismatch error when a maximum mismatch

compensation is applied via the analog compensation circuit **602**. In process block **676**, an IQ mismatch correction scale is applied. The IQ mismatch correction scale may be determined by measuring IQ mismatch as described with respect to the process block **672**, when no compensation is applied, and measuring IQ mismatch as described with respect to the process block **674**, when a maximum compensation is applied to determine an IQ mismatch delta. In process block **678**, the processor **12** may apply an updated IQ mismatch compensation based on the determined IQ mismatch correction scale. In this manner, the method **670** determines an analog IQ mismatch compensation error and applies analog compensation correction to mitigate or eliminate the IQ mismatch compensation error.

(98) FIG. **22** is a flowchart of a method **700** for determining an IQ mismatch and compensating for the IQ mismatch, according to embodiments of the present disclosure. Any suitable device (e.g., a controller, the processor **12**, the measurement and compensation circuitries discussed above, and so on) that may control components of the electronic device **10**, may perform the method **700**. In some embodiments, the method **700** may be implemented by executing instructions stored in a tangible, non-transitory, computer-readable medium, such as the memory **14** and/or storage **16**, using the processor **12**. For example, the method **700** may be performed at least in part by one or more software components, such as an operating system of the electronic device **10**, one or more software applications of the electronic device **10**, and the like. While the method **700** is described using steps in a specific sequence, it should be understood that the present disclosure contemplates that the described steps may be performed in different sequences than the sequence illustrated, and certain described steps may be skipped or not performed altogether.

(99) In process block **702**, the processor **12** may (using the systems and techniques discussed with respect to FIGS. **10-18**, determine an IQ mismatch. In process block **704**, the processor **12** may, via the analog compensation circuit **602**, apply a compensation delay to compensate for the IQ mismatch. In process block **706**, the processor **12** may determine an error in the applied compensation delay, as discussed with respect to the method **670** of FIG. **21**. In process block **708**, the processor **12** may adjust the compensation delay to mitigate the error in the applied compensation delay to improve the IQ mismatch compensation applied via the analog compensation circuit **602**. It should be appreciated that the method **700** may be performed in the background during normal operation of the electronic device **10**. In this manner, the method **700** may determine an IQ mismatch and compensate for the IQ mismatch during normal operation of the electronic device **10**.

(100) The specific embodiments described above have been shown by way of example, and it should be understood that these embodiments may be susceptible to various modifications and alternative forms. It should be further understood that the claims are not intended to be limited to the particular forms disclosed, but rather to cover all modifications, equivalents, and alternatives falling within the spirit and scope of this disclosure.

(101) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . .” or “step for [perform]ing [a function] . . .,” it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

(102) It is well understood that the use of personally identifiable information should follow privacy policies and practices that are generally recognized as meeting or exceeding industry or governmental requirements for maintaining the privacy of users. In particular, personally identifiable information data should be managed and handled so as to minimize risks of unintentional or unauthorized access or use, and the nature of authorized use should be clearly indicated to users.

Claims

1. A digital-to-analog converter, comprising: a delay line comprising a plurality of inverters selectively coupled to an output of the delay line via a first plurality of switches; an additive delay line comprising a second plurality of switches coupled to a plurality of capacitive elements, wherein the additive delay line is coupled between a first inverter and a second inverter of the plurality of inverters; and a latch sensor comprising an input coupled to the output of the delay line and configured to transition a state of a relative signal based on a reference signal overlapping a falling edge of a first quadrature signal at the latch sensor.
2. The digital-to-analog converter of claim 1, wherein the delay line is configured to apply a first delay to the reference signal.
3. The digital-to-analog converter of claim 2, wherein each switch of the first plurality of switches is coupled between two inverters of the plurality of inverters and the output of the delay line.
4. The digital-to-analog converter of claim 3, wherein closing a first switch of the first plurality of switches applies a first adjustment to the first delay and closing a second switch of the first plurality of switches applies a second adjustment to the first delay different from the first adjustment.
5. The digital-to-analog converter of claim 2, wherein the additive delay line is configured to apply a second delay to the reference signal, wherein the second delay comprises a larger delay than the first delay.
6. The digital-to-analog converter of claim 5, wherein closing a first switch of the second plurality of switches applies a first adjustment to the second delay and closing a second switch of the second plurality of switches applies a second adjustment to the second delay.
7. The digital-to-analog converter of claim 1, wherein the latch sensor is configured to transition the state of the relative signal based on the reference signal overlapping a rising edge of the first quadrature signal at the latch sensor, wherein the relative signal is output from the latch sensor.
8. The digital-to-analog converter of claim 1, wherein the reference signal comprises a second quadrature signal.
9. The digital-to-analog converter of claim 1, comprising processing circuitry configured to determine a quadrature signal mismatch between the reference signal and the first quadrature signal based on the transition of the state of the relative signal.
10. A system, comprising: a delay line comprising a plurality of inverters selectively coupled to an output of the delay line via a first plurality of switches, wherein the delay line is configured to apply a first delay to a reference signal; and an additive delay line comprising a second plurality of switches coupled to a plurality of capacitive elements, wherein the additive delay line is coupled between a first inverter and a second inverter of the plurality of inverters; and a latch sensor comprising an input coupled to the output of the delay line, the latch sensor configured to transition a state of a relative signal based on a reference signal overlapping a falling edge of a first quadrature signal at the latch sensor.
11. The system of claim 10, wherein each switch of the first plurality of switches is coupled between two inverters of the plurality of inverters and the output of the delay line, wherein the latch sensor is configured to transition a state of a relative signal based on a reference signal overlapping a rising edge of the first quadrature signal at the latch sensor, wherein the relative signal is output from the latch sensor, wherein the latch sensor is configured to transition the state of the relative signal based on the reference signal overlapping a falling edge of the first quadrature signal at the latch sensor, wherein the reference signal comprises a second quadrature signal.
12. The system of claim 10, wherein closing a first switch of the first plurality of switches applies a first adjustment to the first delay and closing a second switch of the first plurality of switches applies a second adjustment to the first delay different from the first adjustment.

13. The system of claim 10, wherein the additive delay line is configured to apply a second delay to the reference signal, wherein the second delay comprises a larger delay than the first delay.
14. The system of claim 13, wherein closing a first switch of the second plurality of switches applies a first adjustment to the second delay and closing a second switch of the second plurality of switches applies a second adjustment to the second delay.
15. A method, comprising: applying, via a delay line, a first delay to a reference signal, the delay line comprising a plurality of inverters selectively coupled to an output of the delay line via a first plurality of switches; applying, via an additive delay line, a second delay to the reference signal, the additive delay line comprising a second plurality of switches coupled to a plurality of capacitive elements, wherein the additive delay line is coupled between a first inverter and a second inverter of the plurality of inverters; and transitioning, via a latch sensor, a state of a relative signal output from the latch sensor based on the reference signal overlapping a falling edge of a first quadrature signal at the latch sensor, wherein an input of the latch sensor is coupled to the output of the delay line.
16. The method of claim 15, wherein the latch sensor is configured to again transition the state of the relative signal based on the reference signal overlapping a rising edge of the first quadrature signal at the latch sensor.
17. The method of claim 16, wherein the reference signal comprises a second quadrature signal.
18. The method of claim 16, comprising determining, via processing circuitry, a quadrature signal mismatch between the reference signal and the first quadrature signal based on the transition of the state of the relative signal.
19. The digital-to-analog converter of claim 9, wherein the processing circuitry is configured to apply a compensation to the first quadrature signal to compensate for the quadrature signal mismatch by applying a delay to a rising edge of the first quadrature signal or a falling edge of the first quadrature signal.
20. The digital-to-analog converter of claim 1, wherein the relative signal comprises a readback signal output from the latch sensor based on a relationship between the reference signal and the first quadrature signal.
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